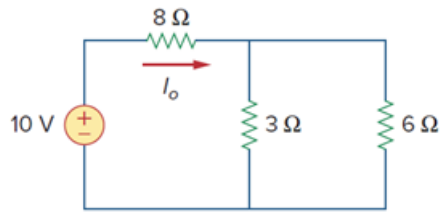


Problem

Calculate I_o in the circuit of Fig. 2.91.

Figure 2.91:



Step-by-step solution

Step 1 of 2

Consider the following circuit diagram:

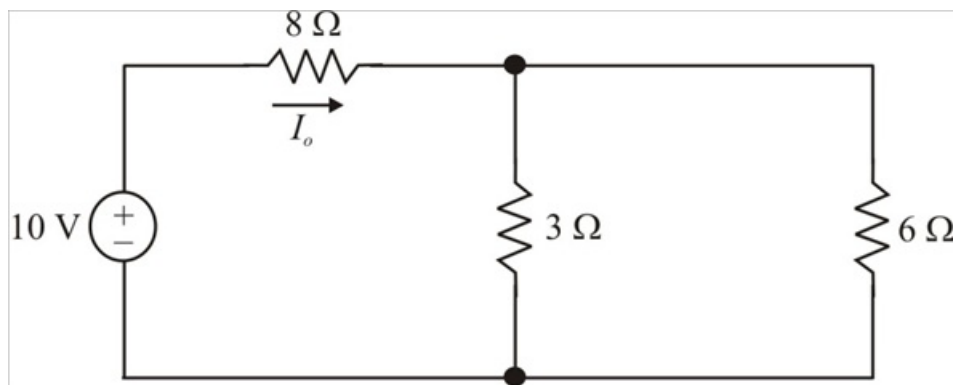


Figure 1

Step 2 of 2

The 3 Ω and 6 Ω resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 3\Omega \parallel 6\Omega &= \frac{3 \times 6}{3 + 6} \\ &= 2\Omega \end{aligned}$$

This 2 Ω and 8 Ω resistors are in series. Their combined resistance is,

$$2\Omega + 8\Omega = 10\Omega$$

By using Ohms law, the current I_o can be calculated as follows.

$$\begin{aligned} I_o &= \frac{v}{R} \\ &= \frac{10}{10} \\ &= 1\text{A} \end{aligned}$$

Therefore, the current I_o is 1A.